

sHū addi Smart Inventory Counter

Eric Jon Smith II

ITAI 1378 – Computer Vision and AI

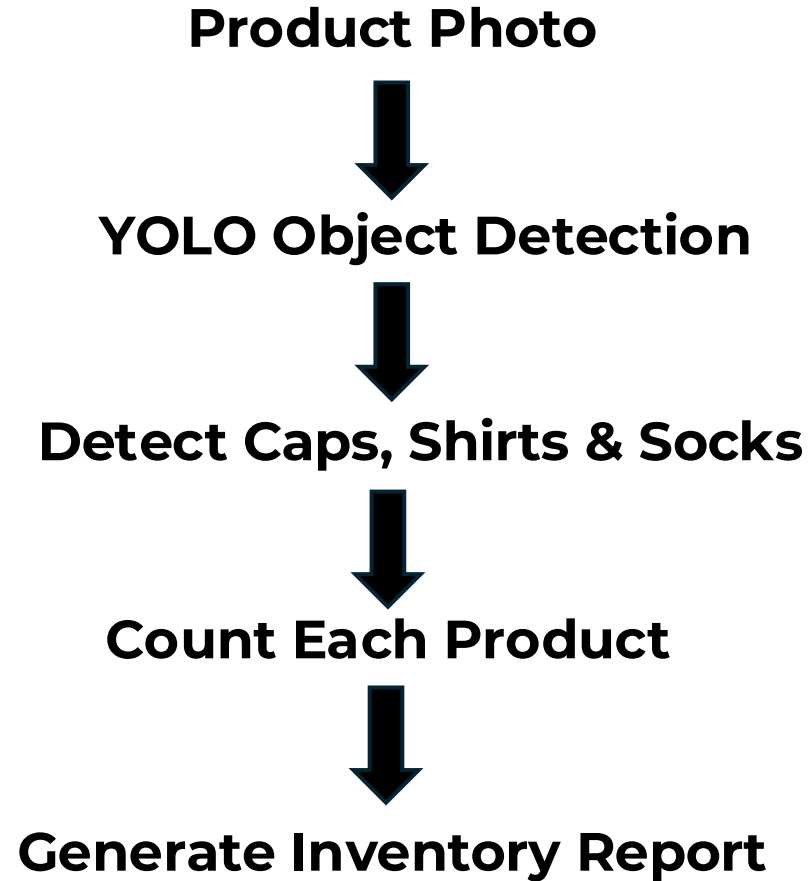
Tier 1: Core

The Problem

Small apparel businesses often count inventory by hand before events, order fulfillment, or restocking. This process takes time and can lead to missed items, incorrect totals, and poor inventory decisions.

sHū addi needs a faster and more reliable way to count products such as caps, shirts, and socks from a single image.

Solution Workflow



Output: The application displays the total number of each product detected and the overall inventory count.

Technical Approach

Computer Vision Technique: Object Detection

Model Architecture: Convolutional Neural Network (CNN)

Model: YOLOv8n

How It Will Be Used: Pretrained model, with optional transfer learning for sHū addi product categories

Framework: Ultralytics YOLO with PyTorch

Why this approach:

YOLOv8n is fast, lightweight, and able to detect multiple objects in one image. It can run on free Google Colab resources and is a practical fit for counting apparel inventory.

Data Plan

Data Source

- Roboflow Universe inventory datasets
- Photos of sHū addi products collected with a smartphone

Estimated Dataset Size

- 300–500 images

Labels

- Hats
- Shirts
- Socks

Label Status

- Public datasets are already labeled.
- Personal photos will be labeled using Roboflow if needed.

Success Metrics

- **Primary Metric**

- Detection Accuracy: Target of 90% or higher

- **Secondary Metrics**

- Inventory counting accuracy of 95% or higher
 - Image processing time of less than 1 second per image

- **Project Goal**

- Demonstrate that the system can accurately detect and count apparel items while reducing the time required for manual inventory counting.

Milestone Plan

Phase	Goal	Timeline
Blueprint	Complete proposal and GitHub repository	WK 5
First Working Demo	Run a pretrained YOLO model on sample images	WK 6
Make it Yours	Test with sHū addi product images	WK 7-8
Improve & Measure	Evaluate accuracy and improve performance	WK 9
Final Build	Complete demo, documentation, and presentation	WK 10

Risk & Resources

Risks

Risk 1: Limited images for testing and training.

Plan B: Use free public datasets from Roboflow Universe and Kaggle until enough sHū addi product images are collected.

Risk 2: Detection accuracy may decrease when products overlap or lighting is poor.

Plan B: Collect additional images with different lighting conditions and camera angles to improve model performance.

Resources

Compute: Google Colab (Free GPU)

Model: YOLOv8n

Framework: Ultralytics YOLO with PyTorch

Dataset: Roboflow Universe + self-collected sHū addi product photos

Cost: \$0 (Free tier and open-source tools only)